

A Packing of 74 Lines in $\mathbb{R}^{13}$

Limin Ge*

October 8, 2026

Abstract

We give 74 1-dimensional subspaces of $\mathbb{R}^{13}$ whose smallest squared chordal distance is at least 0.899931. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 0.899311, obtained by deleting 1 of the 75 lines of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 74 subspaces of dimension 1 in $\mathbb{R}^{13}$ whose pairwise squared chordal distances $D(a, b) = 1 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 0.899931 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 13$, $n = 1$. The cell $N = 74$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 0.899311, obtained by deleting 1 of the 75 lines of the packing found by this search. The Rankin simplex bound for this cell is $1(13 - 1)/13 \cdot 74/73 = 0.935722$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m13n1N74.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 75 seconds on one laptop, with about ten such searches running at once.

A Exact value

```
numerator:
449965578800249980825099534212525717153762904171
denominator:
4999999999999764461798240527203657098467135566349
```

B Coordinates

74 blocks of 13 rows with 1 integers each; divide by 10^{12} . File: `grassmann/packings/m13n1N74.txt`.

149191361837	-84777535067	114433462190	-274551972325
-216865371500	-128620187640	233656833246	-54254552222
219893959935	85961059359	186028806173	65418265189
-351206240090	-158742279972	64355207877	-595713637776
-395602446	-80574728616	563724006424	-42766625782
-248224831609	551752471527	-322087846579	-194164465641
-214271673633	460343018672	210927461028	572718321323
-503681881412	-182277867623	-55287879152	331853695515
130865099690	442901798561	163202665393	52888385429
46532332588	642787228672	311305736532	40089426722
229027695440	-161440499718	-186885620342	-28613619689
402125420456	284781862241	-314887852385	-212781821607
-405387701889	198151969324	-211640919978	471926782684
-339620416311	-56979369856	451594226267	221348414511
-98364490306	-37040967883	-141081482274	193720951310
563598736898	267153085454	22929290941	106909051329
391209393780	-44999930157	184811027918	643765870891
318900366774	324686628701	264825573875	-27622036414
-190638967516	-143846473964	52315581017	-179845370349
117205533119	-100056590330	-45810270495	-202020649085
-145093932950	-173480966276	-343282005338	376934038768
269653432568	50709396753	-21159838888	-28389312061
130579408632	635512090399	123409482661	16107522033
-72924029578	-170491782698	-32313628024	50573581437
-165769848621	-604638269891	-5659002202	479732376670
-330003810831	202698594857	539994595425	-54096614588
-249676702581	78290339241	-242282635806	-224009136927
-9194456542	279558514477	95267155300	418869205709
-163765578229	-156833490895	630007586984	-291008870107
-404247073495	24133261455	-140987432813	-70679169963
-132146136904	100323908514	24476752146	-160459962916
-145536065333	-83997038906	-447910528927	300199854968
375264468248	176165604470	-2550229625	164920133760
122402177436	13968769598	116594277174	-136724651740
-398691165635	314913586883	271709200630	-439576260698
-468636687878	-274649011562	-306328898230	200085873030
-357940527167	225448560492	-169870990697	237898952309
210274442132	307261209495	-398731310865	170059490188
-44512137110	64695172829	237781471090	203932118062
356017668298	274254849814	354060671291	-31822557492
416392372644	101086086469	276401395487	-121517474019
283874451135	206994551443	-389224931898	-43492314883
77156923853	-154662265791	-185761919527	566436028511
187268025053	609618196212	709306649279	114807170763
110047491005	-193772412490	441126017213	244473124478
-127593237323	244589592444	-20818234857	268408039023
-154510740649	-234025514702	17814984690	164483903064
-336455846092	386450210738	-127517976461	277942739541
-271671400364	111744495977	70261577296	-289106819698
-238421166439	296731278446	191799065402	-508025972784
-193842725897	398583369507	238764664658	-513549832295
494540199734	78295768147	-6047687160	169749605436
-27879194526	-663722707672	306145876797	227820374959
104877066959	-19349621746	-236703359199	-250927289230
170364973455	-136722477474	47437623244	317599919881
-488067140850	-53235368740	-117279931799	-54705757755
-198554812669	-99178447014	-421294411866	-73465292803
429166515525	263662843145	-8135241817	-237112325505
301893287344	-198529722042	373838446250	310531864587
46952745603	62635694771	125800665339	192676421318
15290556688	122782933278	510193904348	-283580727196
-58150229191	-144509150913	321860824860	-71190460539
189620028660	814562116166	-358295125216	456139044065
-403460584900	-53287495175	28358724468	-358171524312
449094995652	17625494159	79798717999	390418927904
-415913939316	371934341352	344148688373	-10212289610
-49611355437	21403945995	-36129442952	418426554453
384299709745	-95263009444	-184189973405	484102297192
239138633200	213519196443	49011617432	-101339271128
-85428126547	40296143772	277913969198	-435574019544

-55142883722	-385839362263	-188667265116	-410321871878
-67245855852	357147670527	306843259785	13037393286
-145999042269	-373968989421	361900713787	-74878810348
-92221116038	-245633025998	-162746905383	207270223451
-262892086096	16251127589	446748953135	202905594980
36760147231	-271130015972	422793970710	514943671254
-274356014774	275450440768	72391228312	541751690251
151268754672	-87808961053	-140246391618	240792778036
163412634356	311239781297	140388989316	-321723327742
-10953875750	-300974777452	415090801942	82752502988
314740926553	244341785640	-278120392490	58670510070
235893719339	-64378456913	-164532726369	41367054899
-18722002255	-271033178203	-36594443286	-257420136903
-516845802565	-283012248453	-80947202215	-83200902691
-207928391996	-26741757839	-369067792945	141826611665
-470247287005	98828027711	-307109389172	-129053386933
-62586934963	-533886369649	-34510031556	25246554015
428972856811	78424674597	409700612140	-548592938750
-23097758439	-455786367578	-444712953629	-529649398953
341816265449	-292444540954	-151613437587	139384044733
37925270126	14469385936	340348586058	-32635535547
350506487858	260242118958	-171361689461	197259148598
16787855458	-410764911929	349507432835	225457555543
86193410664	-94779724623	270089089262	-284598401816
-284731452737	-66225409946	-136483783406	-341601098105
61289362399	-188594142074	140480416127	-28909154067
453123728034	383310523654	217072461727	-142372931201
427856237860	-378470054629	-449095409003	17129801212
68539410450	115903221147	122559865473	-92755802244
161748388938	487178798984	-56605038794	-336668484850
261288822896	-253875612292	-435413978593	-92662305337
-422865084884	207091314427	-364118890013	-641717025141
212664005393	234099335845	-105462529006	-48636617832
-439421509287	-107350167128	313346120147	-340175039583
16427921804	-386645624441	278473286736	-465471957126
279985358218	-204279769002	335609469570	-136170655740
202388185776	-60037150861	43114988244	213261404316
-172574034537	327406501519	282700318255	-193630662292
241998192411	-95612293761	255163884994	-53563068710
-169351229339	244138854776	68576551221	-59835855689
474704595099	-484294005654	660598815808	300551629919
61461042137	-313980935061	-194559145031	97423961152
50593757522	165573437785	-40898546336	-729794857049
489061027274	195167755058	-144250903153	-495248786851
232750966488	-82415616261	-55538307141	-136984485189
-23824454477	420949858379	353026575874	-46819553081
56354353061	215891465197	222914686077	210272610866
87812286385	487978288743	411904435703	101323583715
-192304984528	-9348393926	67946431073	-26380440038
244129139862	-434175842734	145160075322	199146281905
-422719866743	166124699907	284307982851	7939451798
-136474016494	239663534663	-246889221850	258731482356
-410377756124	-22999367194	-248059561688	17526319515
-21598699798	-322806758741	331994756388	-222477666108
-299685264088	-451908898164	-242926966194	-341893811369
-532695168938	59095271305	-143552209710	-226194726514
-219120796789	-364702826031	90810953209	-12980513603
-322772934179	167611075639	-132835220498	-196767396503
251296594247	-22426674786	236695692222	107089937355
350857778874	119136273320	376210617522	-27786335936
61307565590	-237979071410	-381282660320	244212048717
-16783133259	-18213326517	465769097066	-429359497181
-206707743407	508299901977	-242068896566	99721032744
256538467148	-202244300111	-163534728598	-640421329714
-156461430344	498675377077	-456117653831	-77505462916
30045466978	-284696762484	430505622806	198411717278
45077487451	-44205329874	-188656549446	-134936334510
105227726302	314705384295	-14855473893	154736470073
19778702607	-267428693691	96510142791	278817642842
807858680973	-351043863798	-13916013418	39485839968
-98254079930	-28168701344	-22109945172	597174446041
-230439184777	-78095829368	-563353382003	153235207545
-76070969405	106758539920	146127592347	-155657575515
-368175595612	-121922611718	356503023423	32979632642
-206272330683	64188627307	98271221451	649168749
87143508053	404738130278	-109637844757	317681963459
205242290947	-613389463505	-471002960086	-571697960866
-430599027631	-186549048806	345260677456	123656967971
-383293548320	99790887215	861370734	291654062314
348700558347	-88923195947	-6369616515	-279991173196
66687388823	140073603126	-315735793623	138858559639
-9116429941	293298341886	133199847435	515257834143
-433038647337	-391932131084	-116221311514	-367394711953
-259454666082	-105885704651	-108920454285	317657141057
-82359585712	-55914935117	576922691181	178118901816
47569686593	-338667116557	294825629852	142926604934
164973169747	84798175496	-85959538153	-371411002259
-483458468093	-513497833043	277373701090	-125266526788
-48214548296	-77638224475	53842755960	134605691858
-284599203792	-645685537575	395574849847	276918130643
58437577586	140265774497	-298564216344	-580584489723
139681754812	119539924487	248037869427	167491797737
268897907311	-152890216331	-352242073456	-175142969785
13083212187	300620967045	-143793793416	313479392759
427213809019	-230213117345	-517017553096	-306445842481
416562946802	33349473945	253006119159	-46139635624
441865264104	168494053465	170445095968	-3287477974
504449977701	-252363916656	-117382313169	-64540215689
151216190658	-110864841804	411669737714	50296137350
-266762647336	240188456933	39200839697	434023377029
32086965595	244604279722	-6994498028	262790895198
-128445303890	3618752900	-97862898162	-378926433532

46468282336	349738481406	-405999479397	-348247750027
-648081655622	268282231483	86761292820	-69600100571
431026734851	53752540354	405709720428	-299710805287
-95352758574	58080078332	220819197904	39116406227
-137722237657	352840107719	187640152123	104825981281
-148358679247	-439192876289	346097175039	413731615571
-166217252587	225957784196	-257483613672	104831827112
-221278975055	456476867123	-292841652799	-360494655376
-70223989226	-389581977356	-598768780378	-161649044366
87630974652	-182179106595	-93506147130	506895601185
10703533465	-421523611859	-15032568003	263517209076
-189440836568	10963379516	-17442100172	275338493220
313393827542	-192325961405	66002735134	27259661
-347440666731	-42524917125	96135319697	203415887308
-357881396914	-140072731670	415185411247	340339098588
287507709307	-50695858052	86916659618	111986960122
282507963605	492372608815	50126025839	-101750509798
10414917119	-420557730172	-100803410476	109171281906
-390546525817	-81138034514	523395223527	69130480199
212207404146	275639687220	136139709679	-98968289449
-145449876927	117805662962	-231308246996	-116969255821
206573195462	465889243909	-157207157198	222325168519
36702613387	168577563208	-156063512747	-819655684459
-374115410111	-75188245698	-297035421859	270292563660
-153910877840	363841721334	-72026362390	67390122945
258160612139	-307533547210	7358653944	-49003231535
465090741641	-40242247585	-207861839050	-299952080068
75204825736	-445839802342	143405484815	213765867267
71098537395	-102082771346	-14411497204	686392708570
287268395245	463533914367	777965606465	377729229090
1594588128	401026136481	-339570659848	161277617437
-159140756851	365057909033	-30041457968	-225345955800
-148861164544	-52234179224	-205424825983	-356873583303
552992148605	-37628976061	-270992019988	-23651231359
-239307186801	207022985847	-196357341055	-86013697898
241629119756	24895621562	329490121985	-168157464634
-513036231465	201512593824	-188643220605	343458980237
348774860853	-130575448812	-323699482144	-87981603304
183250945762	204031405088	360926062369	92706207485
-139263299824	-317615525472	-64430025031	-104524809693
241684443498	552195685433	166143420670	24005722581
54569467255	-93413174773	312282945422	-63087229369
176443599984	-11949156740	-573104933314	167623608952
624579553438	-144064205858	-93329957823	327323264801
387271021020	545723788537	-13173086490	319900031456
335464662188	-183660822102	160474290214	-254986551089
-82174782419	260375920509	-63481438442	310839387540
229838752358	-252328707887	-263728808573	-301415610339
58240225018	47746923159	482769392311	-353537418445
-297641099772	-119572667866	7140222175	163375179066
139039866343	-332227880212	-52277530935	-460367370653
285851429435	-38451880921	208988493291	174705982965
-47535649674	183198626550	45118117308	-309096170912
-488403986942	306535907445	620871045856	-127570554887
269127853174	292618898672	-258865110310	-107710690438
-25380622789	-197419391143	65030077345	277794672870
26708486095	-84089621335	-352055929872	162497878348
-46445332728	273510487339	-196058386823	146123782609
260749812525	-493154014496	207914247374	-353486417938
513380897648	-450859893044	119604943364	228135627189
-239102813418	-160539411123	336132534669	-264892509864
191403309264	259989906703	364825655396	125592775518
-227706420115	220848512087	-524611127130	480136417990
-268406479960	-209267538526	164378386817	250645045435
-331826004817	-141570413668	477859961534	-487777287516
-161742119772	-376261319968	-35926352877	-252032888910
-15923853762	227738591021	-92062439335	-75693655510
19365966514	387084219193	170038691288	
192453837082	-58331249228	31563118739	